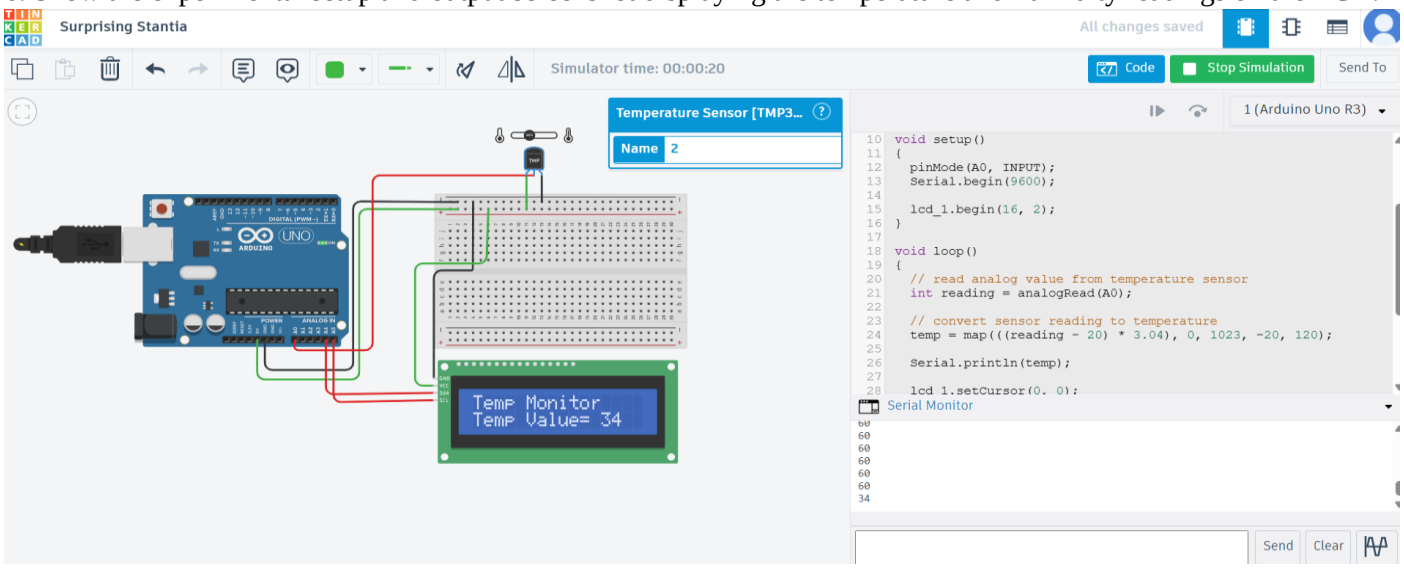
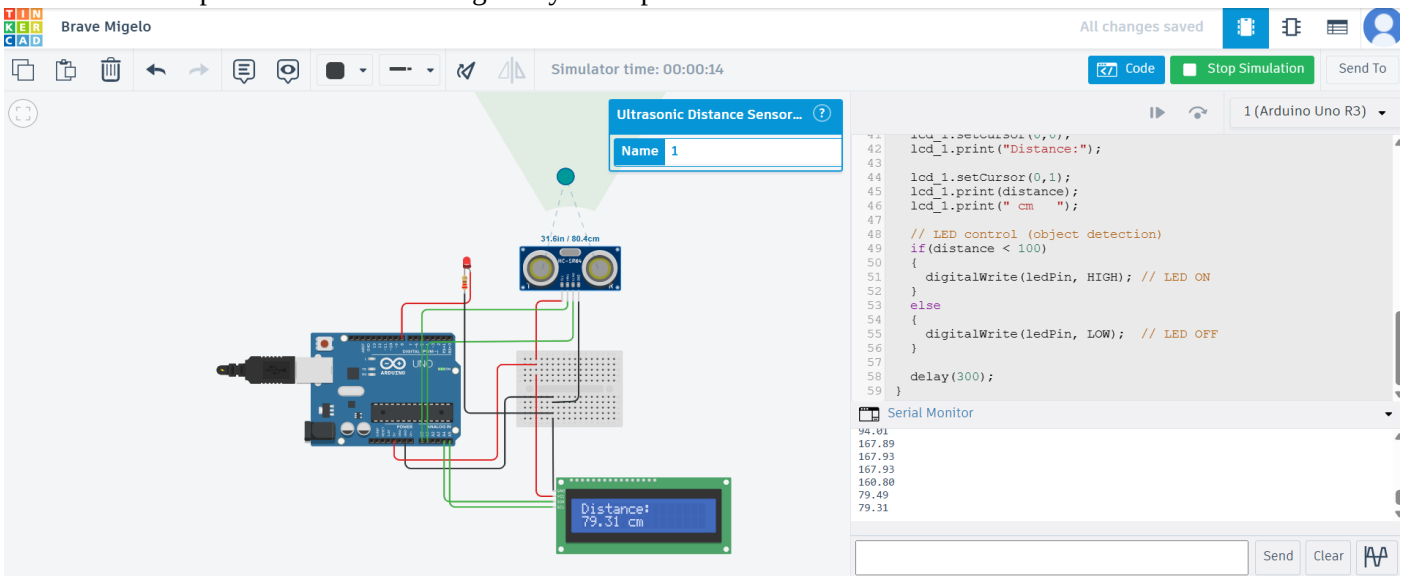
ASSIGNMENT – 9

6. Show the experimental setup and output screenshot displaying the temperature and humidity readings on the LCD.



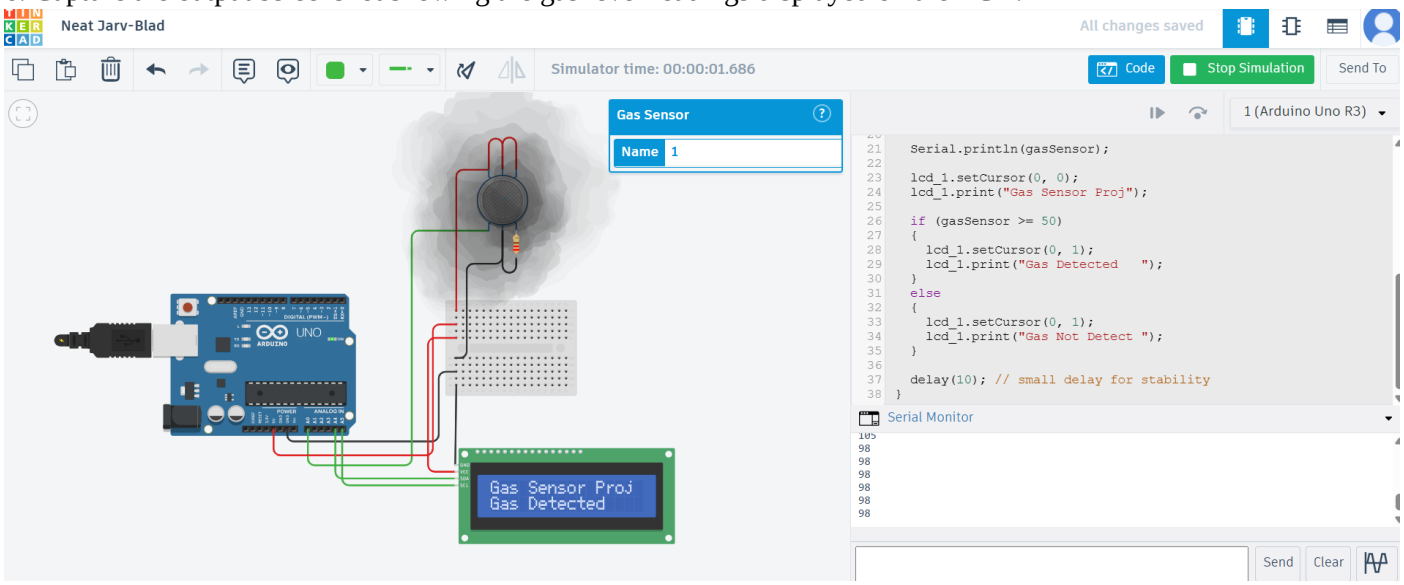
ASSIGNMENT - 10

6. Display the sensor detection status on the LCD.
7. Attach the output screenshots showing the system operation.



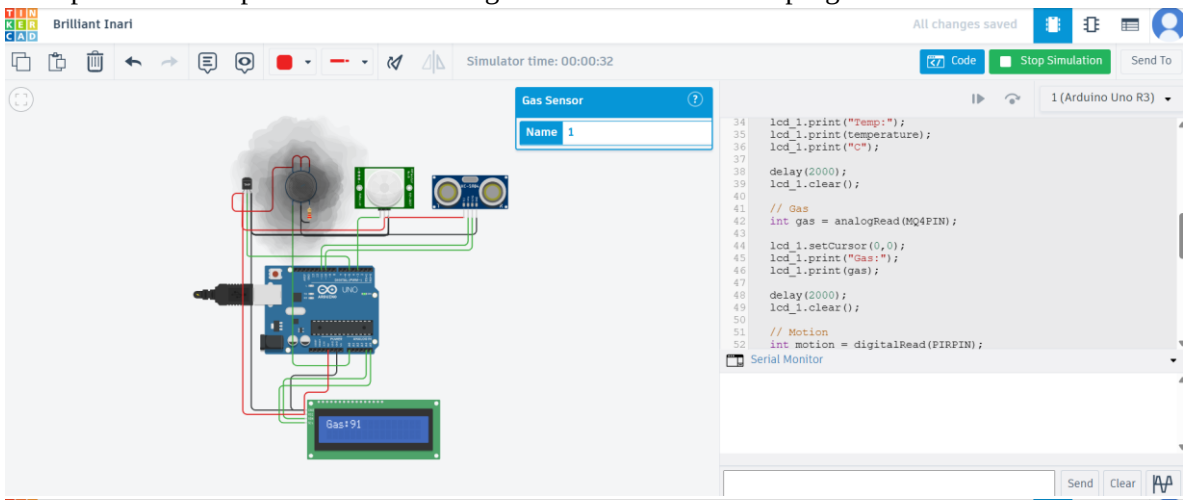
ASSIGNMENT – 11

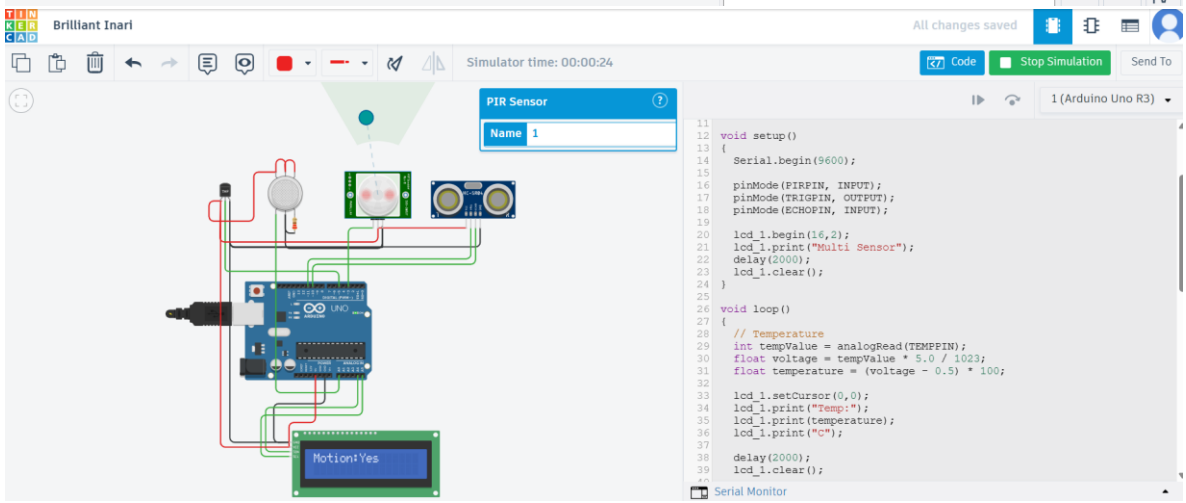
6. Capture the output screenshot showing the gas level readings displayed on the LCD.

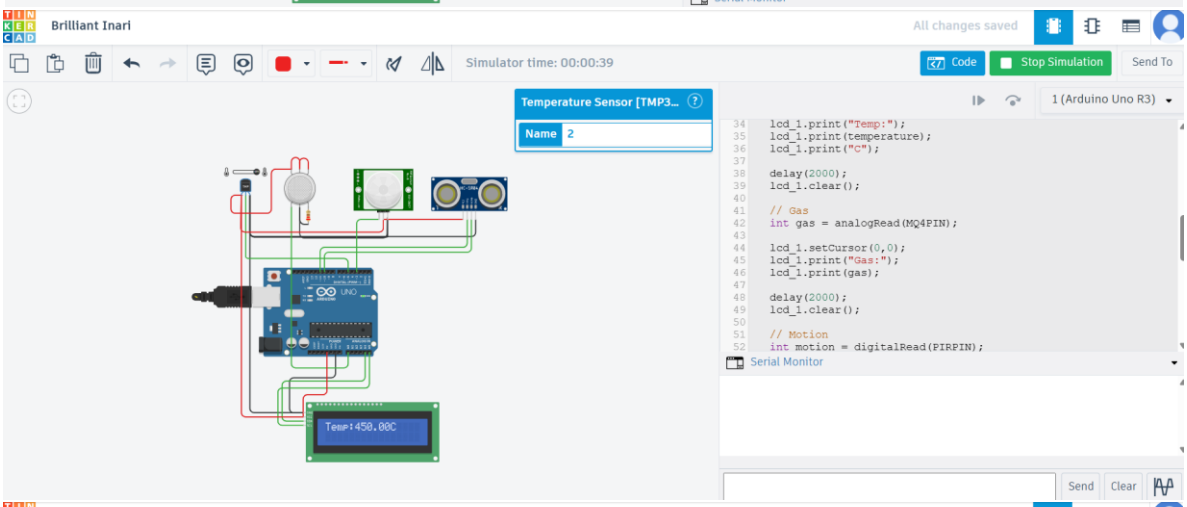


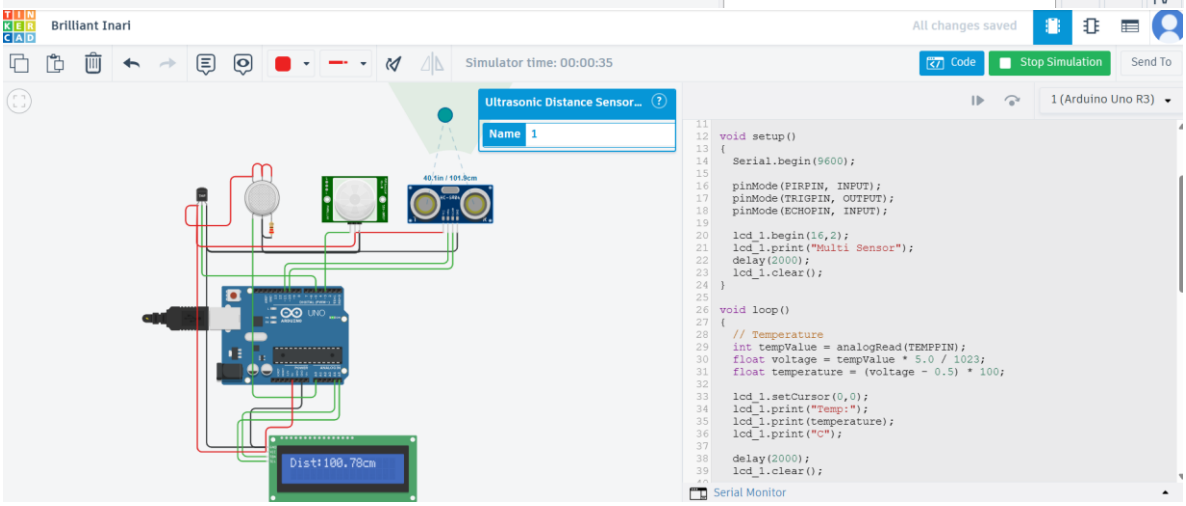
ASSIGNMENT – 12

6. Show the experimental output with LCD readings and screenshots of the program execution.







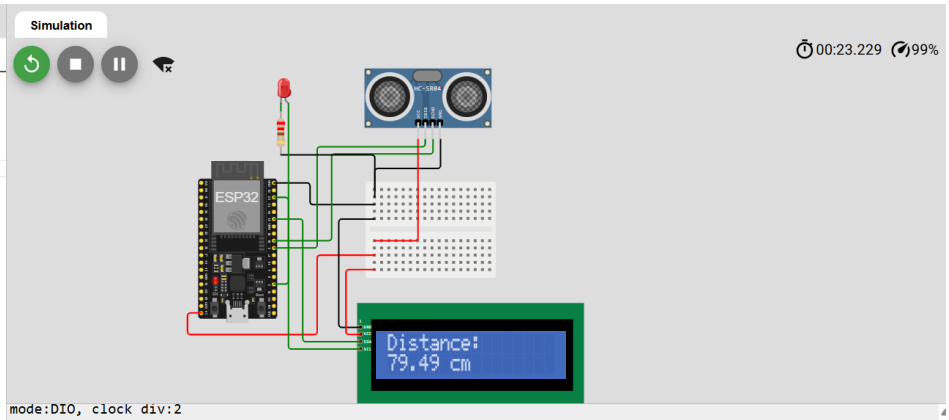


ASSIGNMENT – 13

6. Display the sensor readings on the LCD display and the cloud platform where we can see the data remotely.
7. Attach the screenshots of the cloud dashboard and experimental output.

sketch.ino ● diagram.json ● Library Manager libraries.txt

```
#include <WiFi.h>
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

const char* ssid = "FTTH-5G";
const char* password = "pass@123";

String apiKey = "GMDCNTRIG6F51LKR";
const char* server = "api.thingspeak.com";

WiFiClient client;

#define TRIG 5
#define ECHO 18

LiquidCrystal_I2C lcd(0x27,16,2);

long duration;
float distance;

void setup()
{
  Serial.begin(115200);

  pinMode(TRIG, OUTPUT);
  pinMode(ECHO, INPUT);

  lcd.init();
  lcd.backlight();

  WiFi.begin(ssid, password);
```



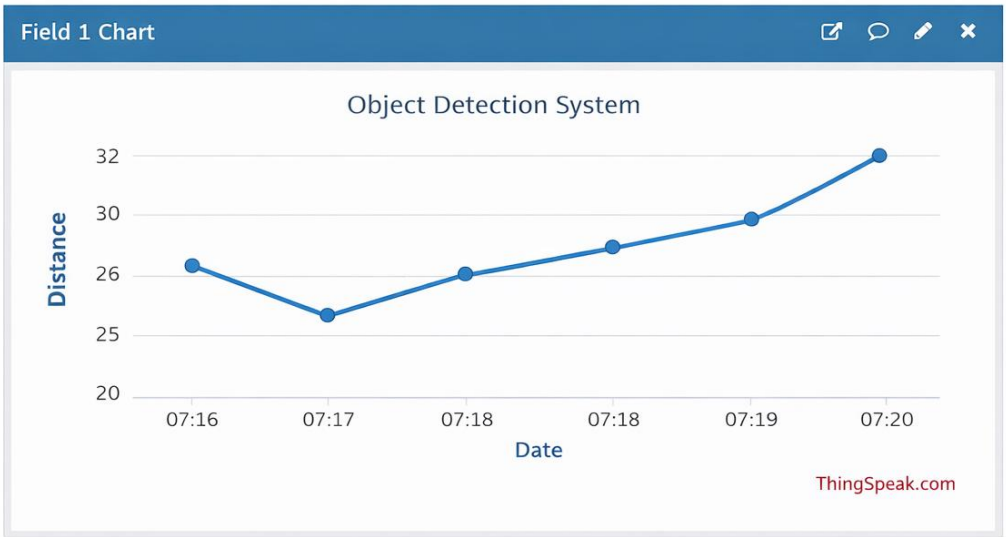

```
mode:DIO, clock div:2
load:0x3fff0030,len:1156
load:0x40078000,len:11456
ho 0 tail 12 room 4
load:0x40080400,len:2972
entry 0x400805dc
.....................................
```

ThingSpeak™ Channels Apps Devices Support Commercial Use How to Buy

Channel Stats

Created: about an hour ago

Entries: 60

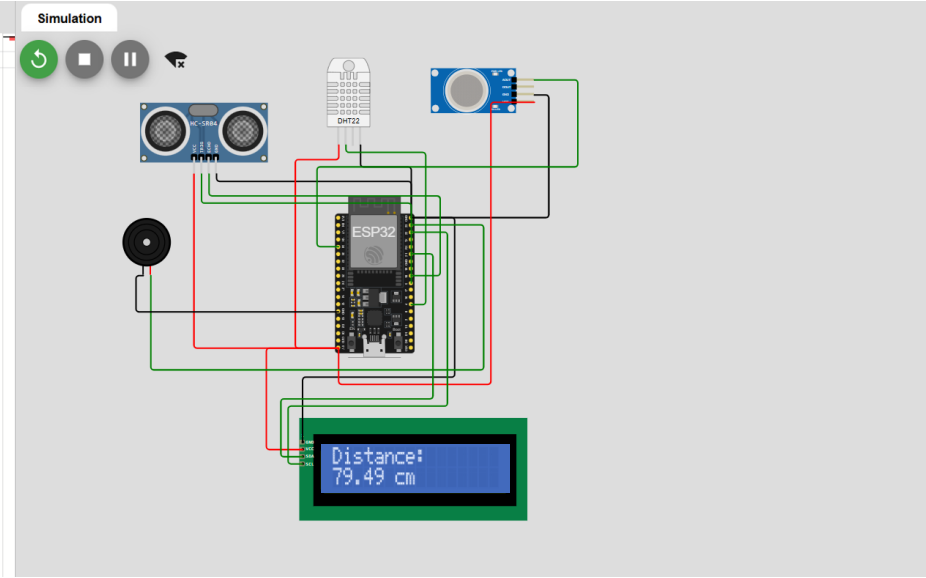


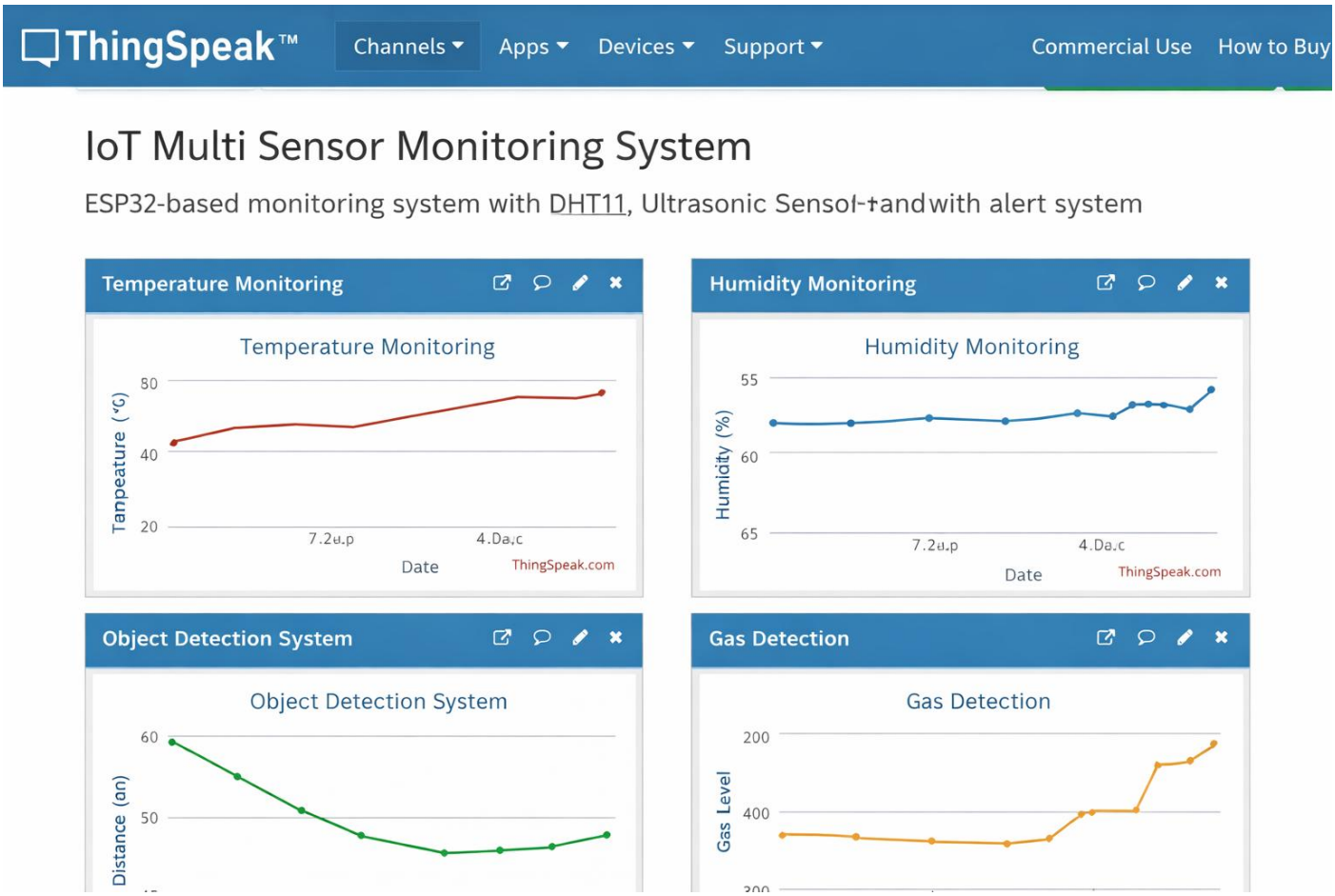
ASSIGNMENT – 14

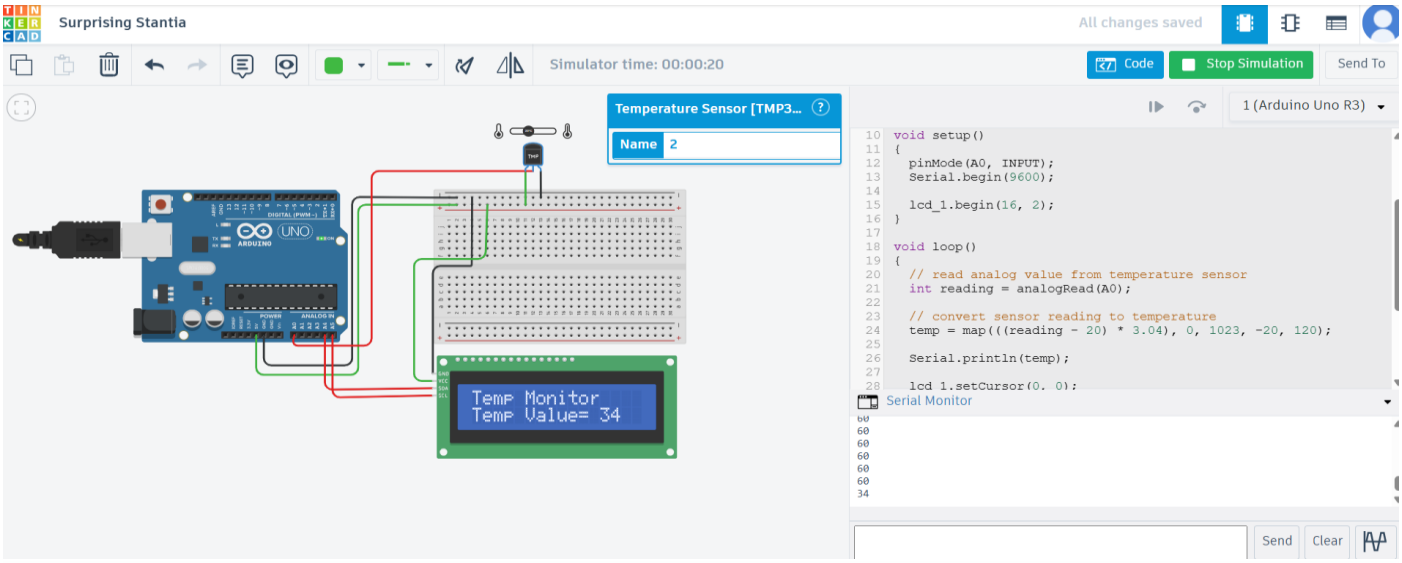
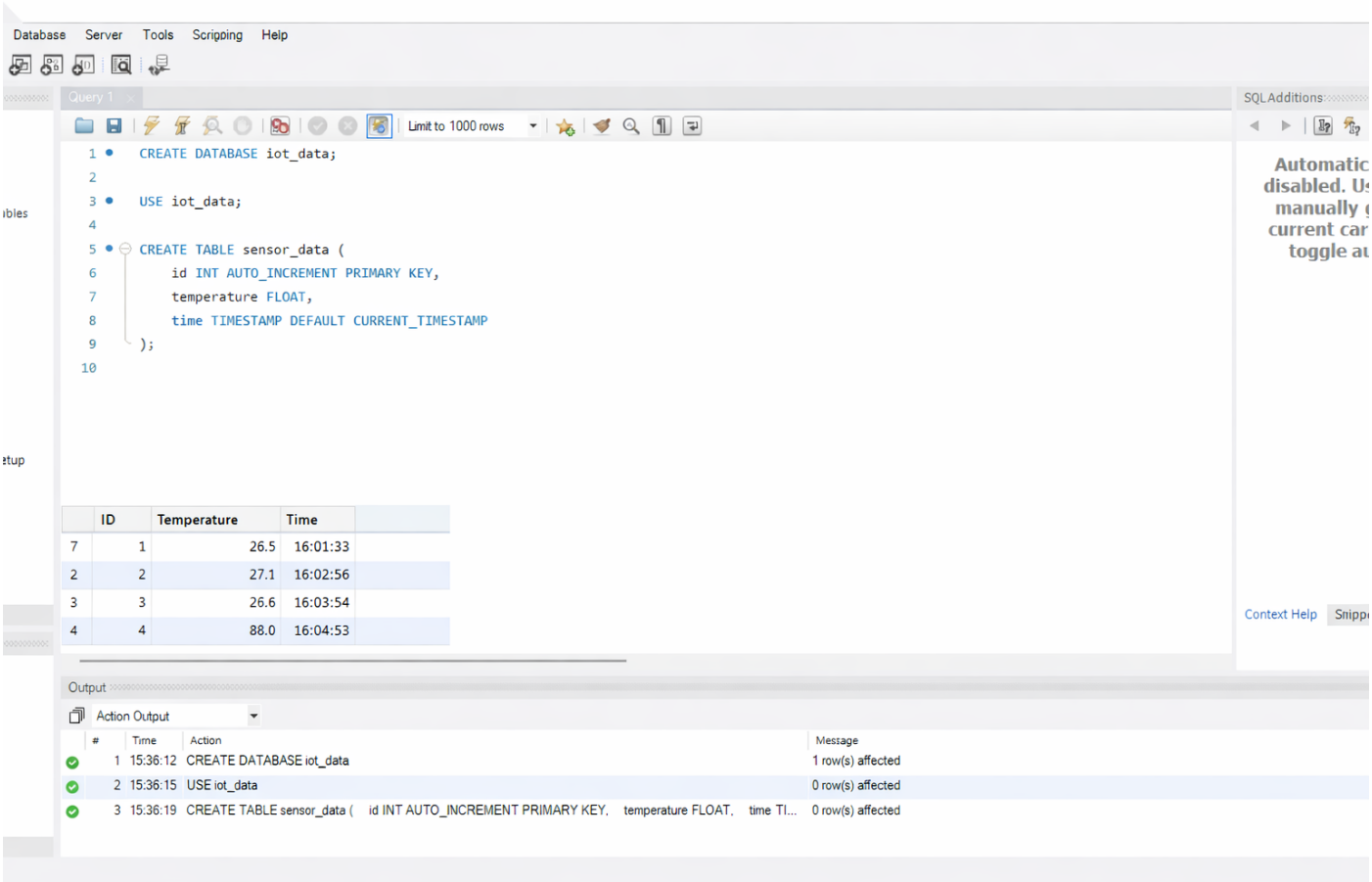
6. Implement an alert system (buzzer/LED/cloud notification) when sensor values exceed predefined thresholds.
7. Provide screenshots of the cloud dashboard, LCD output, and system operation.

sketch.ino ● diagram.json ● Library Manager

```
#include <WiFi.h>
#include <ThingSpeak.h>
#include <DHT.h>
#include <LiquidCrystal_I2C.h>

#define DHTPIN 4
#define DHTTYPE DHT11

#define TRIG 5
#define ECHO 18

#define GAS_SENSOR 34
#define BUZZER 23
#define LED 2

DHT dht(DHTPIN, DHTTYPE);

LiquidCrystal_I2C lcd(0x27,16,2);

const char* ssid = "YOUR_WIFI_NAME";
const char* password = "YOUR_WIFI_PASSWORD";

unsigned long myChannelNumber = YOUR_CHANNEL_ID;
const char * myWriteAPIKey = "YOUR_API_KEY";

WiFiClient client;

void setup()
{
Serial.begin(115200);
```





ASSIGNMENT – 15

7. Configure the private cloud server to receive, store, and display the uploaded sensor data.
8. Demonstrate the working system and provide screenshots of sensor readings, data transmission, and data storage on the private cloud server.